

MATCH IPL

SUBMITTED TO -
MR. LOKESH PAL

WINNER PREDICTION

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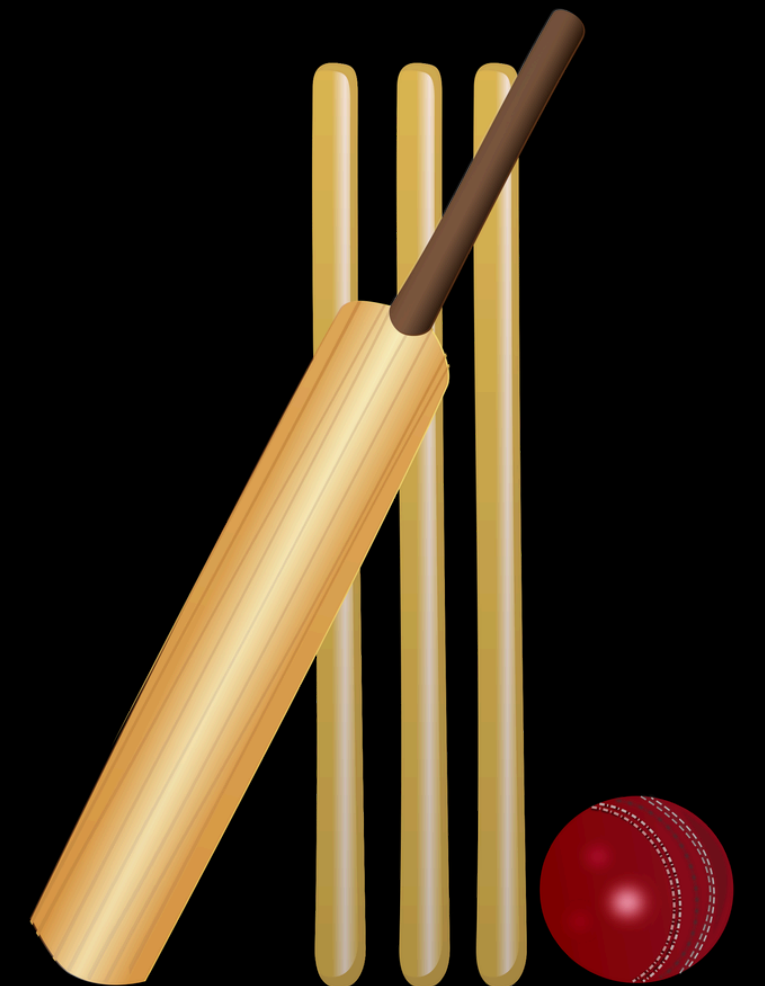
INTRODUCTION

- INDIAN PREMIER LEAGUE (IPL) IS A POPULAR T20 CRICKET LEAGUE
- MATCH OUTCOMES DEPEND ON MANY FACTORS
- PREDICTING RESULTS IS CHALLENGING
- MACHINE LEARNING HELPS ANALYZE PATTERNS FROM PAST DATA

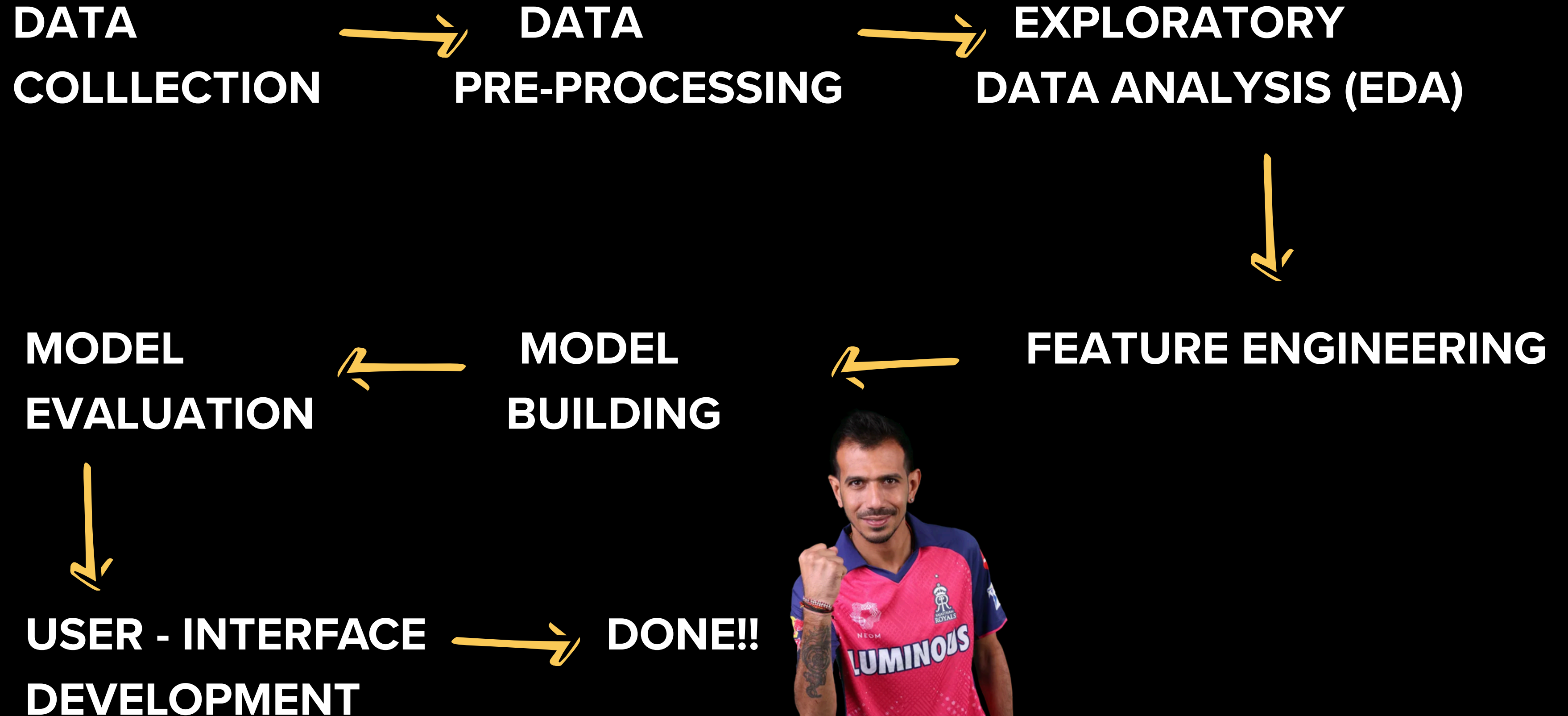


OBJECTIVE

- **PREDICT IPL MATCH WINNERS**
- **ANALYZE HISTORICAL MATCH DATA**
- **BUILD A MACHINE LEARNING MODEL**
- **DEVELOP AN INTERACTIVE WEB APPLICATION**

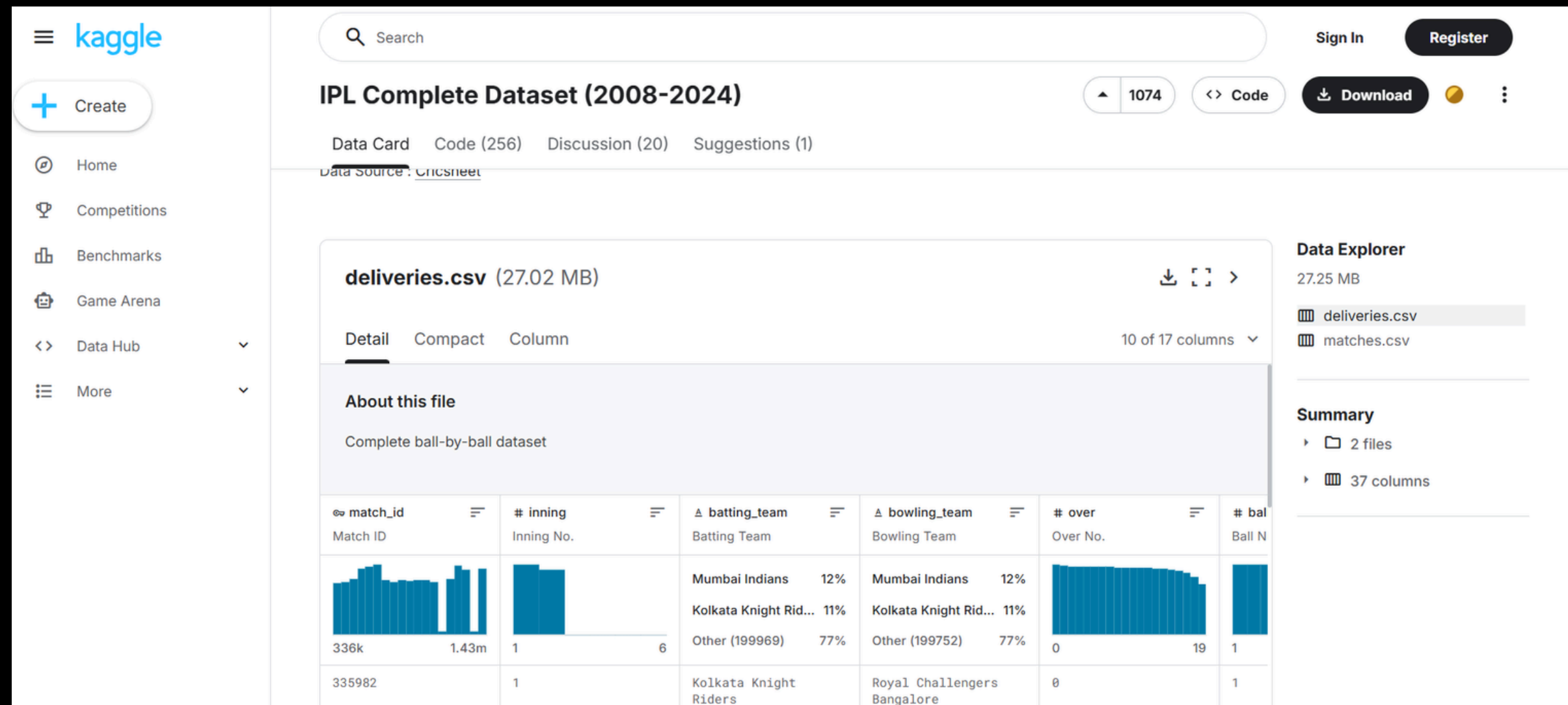


WORKFLOW OF PROJECT!!



DATASET

- DATASET USED: MATCHES.CSV
- CONTAINS HISTORICAL IPL MATCH DATA
- KEY FEATURES :
 - TEAM1, TEAM2
 - TOSS WINNER
 - TOSS DECISION
 - VENUE
 - MATCH WINNER



DATA PREPROCESSING

- **SELECTED RELEVANT COLUMNS**
- **REMOVED MISSING VALUES**
- **CLEANED DATASET FOR ANALYSIS**
- **ENSURED DATA CONSISTENCY**

```
# Selecting relevant columns
matches = matches[['team1', 'team2', 'toss_winner',
                    'toss_decision', 'venue', 'winner']]

# Removing missing values
matches.dropna(inplace=True)
```

EDA OVERVIEW

- EXPLORED DATA TO UNDERSTAND PATTERNS
- USED STATISTICAL SUMMARIES
- CREATED VISUALIZATIONS
- IDENTIFIED KEY TRENDS

```
matches.info()  
matches.describe()
```


VISUALIZATIONS

- VISUALIZED MATCH TRENDS USING GRAPHS
- COMPARED TEAM PERFORMANCES
- ANALYZED TOSS DECISIONS
- STUDIED SEASON-WISE MATCH DISTRIBUTION

```
# Top teams by wins  
matches['winner'].value_counts().head(5).plot(kind='bar')
```

```
# Toss decision distribution  
matches['toss_decision'].value_counts().plot(kind='pie', autopct='%1.1f%%')
```


KEY INSIGHTS

- **SOME TEAMS PERFORM CONSISTENTLY BETTER**
- **TOSS WINNER HAS SLIGHT ADVANTAGE**
- **TEAMS PREFER FIELDING FIRST**
- **VENUE PLAYS AN IMPORTANT ROLE**
- **MATCH TRENDS VARY ACROSS SEASONS**

```
matches['winner'].value_counts()
```

FEATURE ENGINEERING

- CREATED NEW FEATURES TO IMPROVE MODEL PERFORMANCE
- TEAM STRENGTH BASED ON PAST WINS
- TOSS WIN FEATURE
- BATTING FIRST FEATURE
- STRENGTH DIFFERENCE BETWEEN TEAMS



```
team_wins = y_train.value_counts().to_dict()

X['team1_strength'] = X['team1'].map(team_wins)
X['team2_strength'] = X['team2'].map(team_wins)

# Toss and batting
X['toss_win'] = (X['toss_winner'] == X['team1']).astype(int)
X['bat_first'] = (X['toss_decision'] == 'bat').astype(int)

# Strength difference
X['strength_diff'] = X['team1_strength'] - X['team2_strength']
```

MODEL USED

- **MODEL USED: RANDOM FOREST CLASSIFIER**
- **IT IS AN ENSEMBLE LEARNING ALGORITHM**
- **WORKS BY COMBINING MULTIPLE DECISION TREES**
- **SUITABLE FOR CLASSIFICATION PROBLEMS**

WHY RANDOM FOREST?

- **HANDLES COMPLEX AND NON-LINEAR DATA**
- **REDUCES OVERFITTING**
- **PROVIDES GOOD ACCURACY**
- **WORKS WELL WITH STRUCTURED DATA**

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=500,
    max_depth=12,
    random_state=42
)
```

MODEL PERFORMANCE

MODEL EVALUATED USING TRAINING AND TESTING DATA

- **TRAINING ACCURACY: ~81%**
- **TESTING ACCURACY: ~47%**

INTERPRETATION:

- **HIGH TRAINING ACCURACY → MODEL LEARNED PATTERNS WELL**
 - **LOWER TESTING ACCURACY → REAL-WORLD PREDICTION IS DIFFICULT**
- SPORTS OUTCOMES ARE UNPREDICTABLE**

USER INTERFACE

localhost:8501

eleceed,read elecee... AI summary ChatGPT PostgreSQL Learnin... All Bookmarks

Deploy



IPL Match Winner Predictor

Predict match outcomes using Machine Learning

Select Team 1

Mumbai Indians

Select Team 2

Royal Challengers Bangalore

Select Venue

Arun Jaitley Stadium

Toss Winner

Mumbai Indians

Toss Decision

bat

 VS 

Predict Winner

RESULT

- PREDICTS MATCH WINNER
- DISPLAYS CONFIDENCE SCORE
- REAL-TIME PREDICTION SYSTEM

 Predict Winner



Royal Challengers Bangalore Wins!

Confidence: 42.41%

CONCLUSION

- DEVELOPED AN IPL MATCH PREDICTION MODEL
- USED HISTORICAL MATCH DATA FOR ANALYSIS
- PERFORMED DATA CLEANING AND PREPROCESSING
- APPLIED EXPLORATORY DATA ANALYSIS (EDA)
- CREATED IMPORTANT FEATURES FOR BETTER PREDICTION
- IMPLEMENTED RANDOM FOREST ALGORITHM
- ACHIEVED DECENT MODEL ACCURACY
- BUILT AN INTERACTIVE UI USING STREAMLIT
- COMPLETED END-TO-END MACHINE LEARNING PROJECT

FUTURE IMPROVEMENTS

- **INCLUDE PLAYER PERFORMANCE STATISTICS**
- **ADD WEATHER AND PITCH CONDITION DATA**
- **USE MORE ADVANCED MACHINE LEARNING MODELS**
- **IMPROVE ACCURACY WITH LARGER DATASETS**
- **ENHANCE UI WITH MORE VISUALIZATIONS**
- **DEVELOP A MOBILE-FRIENDLY OR FULL WEB APPLICATION**



**ANY
QUESTIONS?**